Predictive Machine Learning via Logistic Regression

Clinical Analysis of the Pima Indians Diabetes Dataset

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Roll no: 36

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- Data set overview
- Exploratory Data Analysis
- Model Estimation & Feature Significance
- Model Diagnostics & Risk Sizing
- Model evaluation
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Data Overview

Pima Indians Diabetes Dataset

```
   pregnant         glucose         pressure         triceps          insulin             mass          pedigree            age          diabetes
 Min.   : 0.000   Min.   : 44.0   Min.   : 24.00   Min.   : 7.00   Min.   : 14.00   Min.   :18.20   Min.   :0.0780   Min.   :21.00   neg:500
 1st Qu.: 1.000   1st Qu.: 99.0   1st Qu.: 64.00   1st Qu.:22.00   1st Qu.: 76.25   1st Qu.:27.50   1st Qu.:0.2437   1st Qu.:24.00   pos:268
 Median : 3.000   Median :117.0   Median : 72.00   Median :29.00   Median :125.00   Median :32.30   Median :0.3725   Median :29.00
 Mean   : 3.845   Mean   :121.7   Mean   : 72.41   Mean   :29.15   Mean   :155.55   Mean   :32.46   Mean   :0.4719   Mean   :33.24
 3rd Qu.: 6.000   3rd Qu.:141.0   3rd Qu.: 80.00   3rd Qu.:36.00   3rd Qu.:190.00   3rd Qu.:36.60   3rd Qu.:0.6262   3rd Qu.:41.00
 Max.   :17.000   Max.   :199.0   Max.   :122.00   Max.   :99.00   Max.   :846.00   Max.   :67.10   Max.   :2.4200   Max.   :81.00
                  NAs    :5       NAs    :35       NAs    :227     NAs    :374      NAs    :11
```

```
Rows: 768
Columns: 9
$ pregnant <dbl> 6, 1, 8, 1, 0, 5, 3, 10, 2, 8, 4, 10, 10, 1, 5, 7, 0, 7, 1, 1, 3, 8, 7, 9, 11, 10, 7, 1, 13, 5, 5, 3, 3, 6, 10, 4, 11, 9, 2, 4, 3, 7, 7, …
$ glucose  <dbl> 148, 85, 183, 89, 137, 116, 78, 115, 197, 125, 110, 168, 139, 189, 166, 100, 118, 107, 103, 115, 126, 99, 196, 119, 143, 125, 147, 97, 14…
$ pressure <dbl> 72, 66, 64, 66, 40, 74, 50, NA, 70, 96, 92, 74, 80, 60, 72, NA, 84, 74, 30, 70, 88, 84, 90, 80, 94, 70, 76, 66, 82, 92, 75, 76, 58, 92, 7…
$ triceps  <dbl> 35, 29, NA, 23, 35, NA, 32, NA, 45, NA, NA, NA, NA, 23, 19, NA, 47, NA, 38, 30, 41, NA, NA, 35, 33, 26, NA, 15, 19, NA, 26, 36, 11, NA, 3…
$ insulin  <dbl> NA, NA, NA, 94, 168, NA, 88, NA, 543, NA, NA, NA, NA, 846, 175, NA, 230, NA, 83, 96, 235, NA, NA, NA, 146, 115, NA, 140, 110, NA, NA, 245…
$ mass     <dbl> 33.6, 26.6, 23.3, 28.1, 43.1, 25.6, 31.0, 35.3, 30.5, NA, 37.6, 38.0, 27.1, 30.1, 25.8, 30.0, 45.8, 29.6, 43.3, 34.6, 39.3, 35.4, 39.8, 2…
$ pedigree <dbl> 0.627, 0.351, 0.672, 0.167, 2.288, 0.201, 0.248, 0.134, 0.158, 0.232, 0.191, 0.537, 1.441, 0.398, 0.587, 0.484, 0.551, 0.254, 0.183, 0.52…
$ age      <dbl> 50, 31, 32, 21, 33, 30, 26, 29, 53, 54, 30, 34, 57, 59, 51, 32, 31, 31, 33, 32, 27, 50, 41, 29, 51, 41, 43, 22, 57, 38, 60, 28, 22, 28, 4…
$ diabetes <fct> pos, neg, pos, neg, pos, neg, pos, neg, pos, pos, neg, pos, neg, pos, pos, pos, pos, pos, neg, pos, neg, neg, pos, pos, pos, pos, pos, ne…
```

Exploratory Data Analysis

```
colSums(is.na(PimaIndiansDiabetes2))
```

```
pregnant  glucose pressure  triceps  insulin     mass pedigree      age diabetes
       0        5       35      227      374       11        0        0        0
```

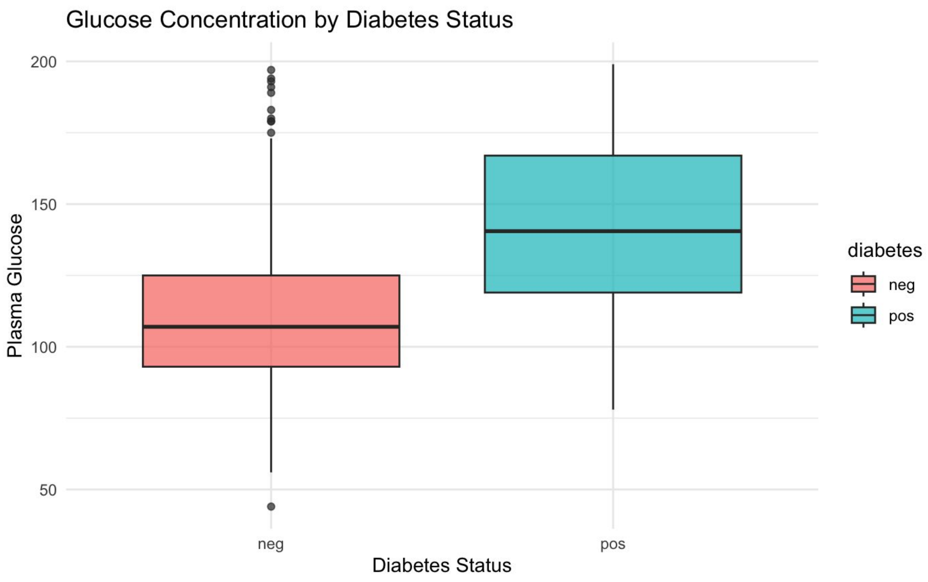
To avoid discarding partial information, used **Multiple Imputation by Chained Equations (MICE)** using **Predictive Mean Matching (PMM)** to fill missing data points based on surrounding patient trends.

```
prop.table(table(clean_diabetes$diabetes))
```

```
      neg       pos
0.6510417 0.3489583
```

Exploratory Data Analysis

The graph provides visual proof that `**glucose**` is a highly significant differentiator, while simultaneously showing enough variance and overlap to confirm that a **multivariable** statistical model is necessary to make accurate predictions.



Model Estimation & Diagnostics

```
diabetes_model <- glm(diabetes ~ ., data = train_set, family = binomial)
summary(diabetes_model)
```

Highly Significant Predictors:

- glucose: By far the strongest predictor.
- Pregnant
- mass (BMI)
- pedigree

Insight:

- Age is not significant here.
- That doesn't mean age doesn't matter in real life; it means that once we already know a patient's glucose, BMI, and pregnancy history, adding their exact age provides no additional predictive power to the model.

```
Call:
glm(formula = diabetes ~ ., family = binomial, data = train_set)

Coefficients:
              Estimate Std. Error z value Pr(>|z|)    
(Intercept) -9.4588942  0.9397847 -10.065  < 2e-16 ***
pregnant     0.1652091  0.0366958   4.502 6.73e-06 ***
glucose      0.0407534  0.0049334   8.261  < 2e-16 ***
pressure    -0.0045708  0.0095603  -0.478    0.633    
triceps      0.0044097  0.0134558   0.328    0.743    
insulin     -0.0008093  0.0011466  -0.706    0.480    
mass         0.0871179  0.0219512   3.969 7.23e-05 ***
pedigree     1.0301626  0.3333315   3.091    0.002 ** 
age          0.0001159  0.0110795   0.010    0.992    
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 796.05  on 614  degrees of freedom
Residual deviance: 557.13  on 606  degrees of freedom
AIC: 575.13

Number of Fisher Scoring iterations: 5
```

Multicollinearity test

```
```{r multicollinearity test}
vif(diabetes_model)

```
```

```
 pregnant   glucose pressure  triceps  insulin     mass pedigree      age 
 1.481160 1.464261 1.237956 1.625870 1.463801 1.816190 1.015777 1.629219 
```

- No VIF metrics cross above the conservative benchmark threshold(>10).
- Structural multicollinearity is not an issue in this context.

Model Evaluation

Accuracy : 0.7647

- **True Negatives (87)**: The model correctly predicted 87 healthy patients as neg.
- **True Positives (30):** The model correctly predicted 30 diabetic patients as pos.
- **False Positives (13):** The model flagged 13 healthy people as diabetic.
- **False Negatives (23)**: The model cleared 23 diabetic patients, telling them they were healthy.

````
```{r confusion matrix}
confusionMatrix(test_set$pred_class, test_set$diabetes, positive = "pos")
```
````

```
Confusion Matrix and Statistics

          Reference
Prediction neg pos
       neg  87  23
       pos  13  30

               Accuracy : 0.7647
                 95% CI : (0.6894, 0.8294)
    No Information Rate : 0.6536
    P-Value [Acc > NIR] : 0.001988

                  Kappa : 0.4563

 Mcnemar's Test P-Value : 0.133614

            Sensitivity : 0.5660
            Specificity : 0.8700
         Pos Pred Value : 0.6977
         Neg Pred Value : 0.7909
             Prevalence : 0.3464
         Detection Rate : 0.1961
   Detection Prevalence : 0.2810
      Balanced Accuracy : 0.7180

       'Positive' Class : pos
```

Model Evaluation

Sensitivity (0.5660):
Out of all the people who actually have diabetes in test pool, the model successfully caught ***56.6%*** of them.

Critique:
This is the weakest link in the current model. It means we are missing ***43.4%*** of the diabetic patients (the 23 False Negatives), sending them home without treatment.

Specificity (0.8700):
Out of all the truly healthy people, the model correctly identified ***87.0%*** of them. It is highly specific, meaning it rarely panics or triggers false alarms for healthy individuals.

````
```{r confusion matrix}
confusionMatrix(test_set$pred_class, test_set$diabetes, positive = "pos")
```
````

```
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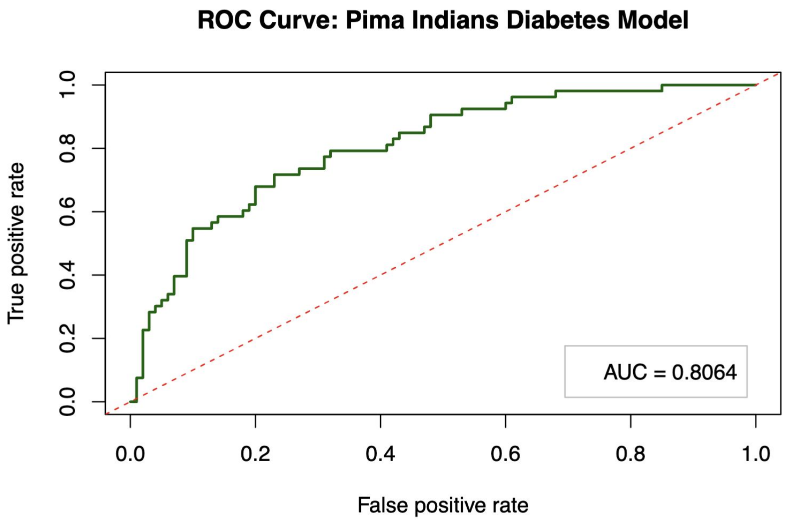
ROC Curve

- **The Probability Concept:** If we randomly select one patient who actually has diabetes and one patient who is completely healthy, and run both through the model, there is an **80.64% chance** that the model will assign a higher risk probability score to the diabetic patient than to the healthy one.
- **Clinical Utility:** This chart proves that if a medical facility wants to catch more diabetic patients (raise Sensitivity), they can use the curve to find the exact mathematical threshold required to do so, while knowing exactly how many false alarms (FPR) they will have to tolerate in return.



Prediction

Table 1: Model Assessment of New Patient Samples

glucose	mass	pedigree	Predicted_Probability	Predicted_Class
165	38.5	1.25	0.9267278	Positive
95	22.0	0.25	0.0283508	Negative

Conclusion

- **Data Preservation via MICE:** Deploying **MICE imputation** with Predictive Mean Matching resolved major missing data gaps (insulin/triceps) without losing sample size. This eliminated selection bias and created a clean dataset with zero missing values.

- **Key Risk Hierarchy: Plasma glucose**, **BMI**, **family pedigree**, and **pregnancy history** are the primary statistically significant drivers of diabetes risk ($p < 0.05$). Genetic pedigree is the heaviest weight, scaling baseline odds by **2.8 times** per unit increase.

- **Robust Global Performance:** The model achieved a **76.47% accuracy** and a **Good discrimination capacity (AUC = 0.8064)** on validation test data. While highly specific at clearing healthy individuals (87%), future clinical workflows should lower the 0.5 decision threshold to improve sensitivity and catch more true positive cases.

References

Packages:

- **mlbench**: For Pima Indians Diabetes 2 data
- **Tidyverse**: For data transformation & plots
- **caret**: For partitioning & confusion matrix
- **ROCR**: For ROC & AUC calculation
- **mice**: For handling missing values via imputation
- **car**: For VIF calculations

Tools used:

- Google Gemini

Thank You

Any queries?